



Unifying Global and Local Features

A Modern Approach to Sports Video Analysis

- 1. Introduction**
- 2. UGL Architecture**
- 3. Benchmark Datasets**
- 4. Quantitative Results**
- 5. Conclusion**

1. Introduction

Task Overview

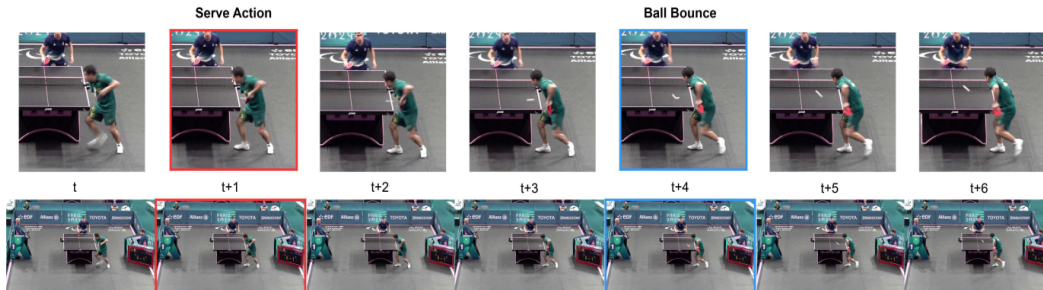


Fig. 3. Example of Precise Event Spotting: in table tennis, the moment a player contacts the ball during a serve (red) or when a ball bounces on the table (blue).

Xu H, Baniya AA, Well S, Bouadjenek MR, Dazeley R, Aryal S. Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges. arXiv:2505.03991, 2025.

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- **Domain:** Action spotting in sports videos (soccer, diving, gymnastics)

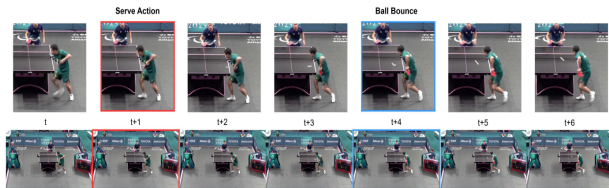


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- **Goal:** Detect *precise timestamps* of events, not long temporal segments

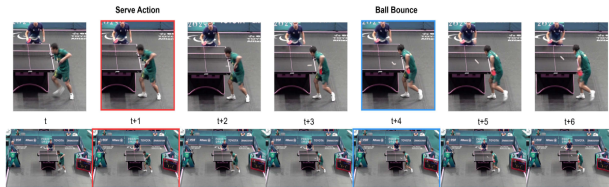


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- **Applications:**
 - ▶ Highlight generation
 - ▶ Video analytics and broadcasting automation
 - ▶ Downstream temporal retrieval

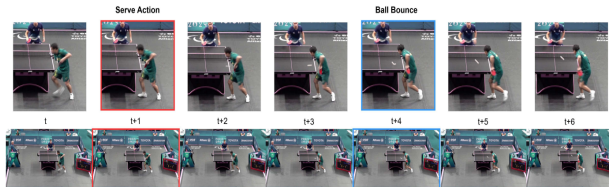


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Motivation for UGL

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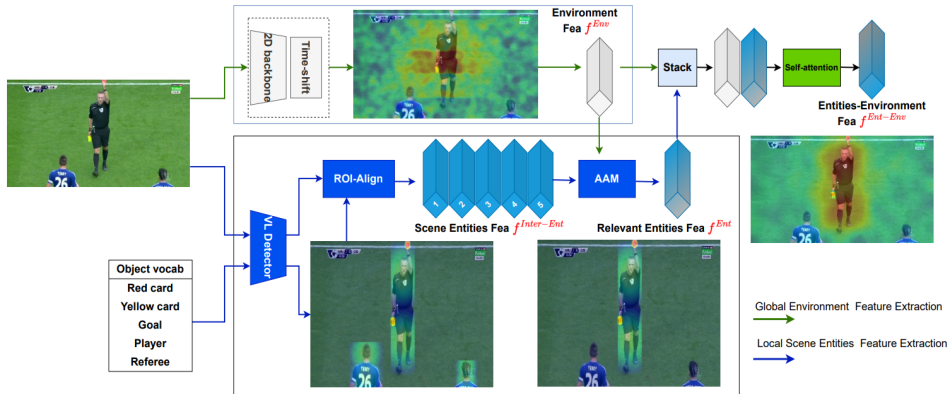
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- **Need**: explicit modeling of both global and local scene elements

2. UGL Architecture

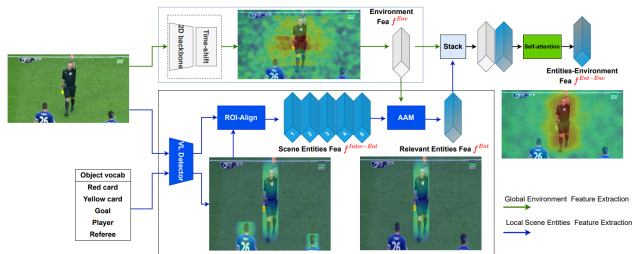
Global Environment Feature



2. UGL Architecture

Global Environment Feature

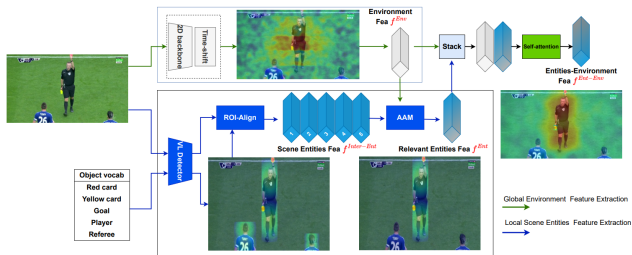
- **Backbone**: RegNet-Y (efficient 2D CNN design)



2. UGL Architecture

Global Environment Feature

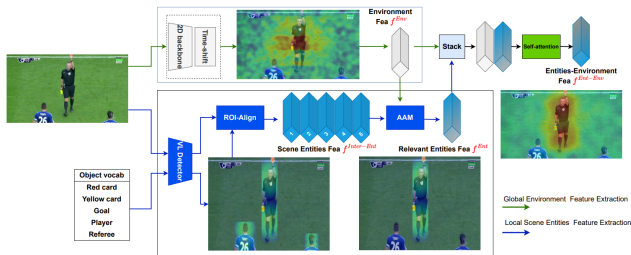
- **Backbone:** RegNet-Y (efficient 2D CNN design)
- **Temporal Integration:** Gate-Shift Module (GSM)
 - ▶ Shifts feature maps forward and backward along temporal axis
 - ▶ Shares information across frames
 - ▶ Lightweight alternative to 3D CNNs



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Global Environment Feature

- **Backbone:** RegNet-Y (efficient 2D CNN design)
- **Temporal Integration:** Gate-Shift Module (GSM)
 - ▶ Shifts feature maps forward and backward along temporal axis
 - ▶ Shares information across frames
 - ▶ Lightweight alternative to 3D CNNs
- **Output:** Global environment feature vector per frame capturing spatial and short-term temporal context



Why GLIP?

Global features alone often ignore small but crucial entities (cards, ball, etc.), especially under class imbalance

- **GLIP**: Grounded Language-Image Pre-training (training-free detector)
- **Input**: Frame image + text prompt listing sports entities
 - ▶ *“ball, yellow card, red card, referee, . . . ”*
- **Mechanism**:
 - ▶ Language encoder processes text (BERT)
 - ▶ Visual encoder produces image proposals (Swin backbone)
 - ▶ Cross-modal multi-head attention aligns proposals with textual entities

2. UGL Architecture

Adaptive Attention Mechanism (AAM)

Problem

Not all detected entities are equally relevant for the current action

AAM Solution

Takes both local entity features and global environment feature:

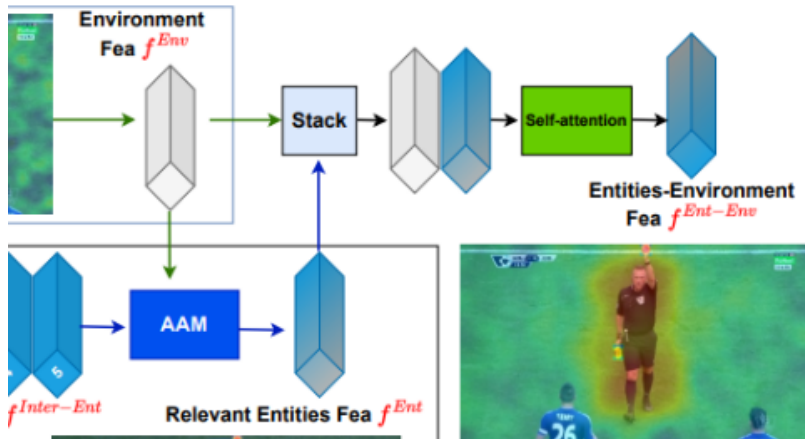
1. **Hard attention**: Select most promising entities
2. **Soft self-attention**: Fuse them into single compact representation

Output

Local relevant scene entities feature emphasizing objects crucial for the action

2. UGL Architecture

Fusion of Global and Local

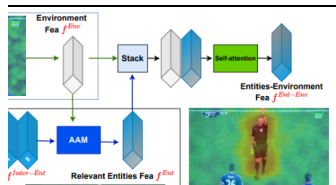


2. UGL Architecture

Fusion of Global and Local

1. Concatenate:

- ▶ Global environment feature
- ▶ Local relevant entities feature



2. UGL Architecture

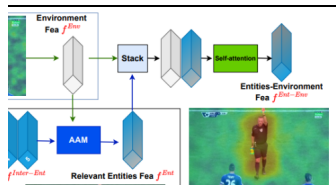
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1. Concatenate:

- ▶ Global environment feature
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2. Apply:

- ▶ Self-attention layer to model interactions
- ▶ Average pooling for compactness



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Fusion of Global and Local

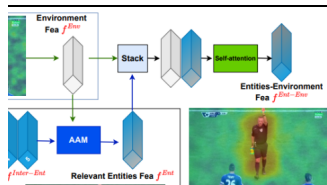
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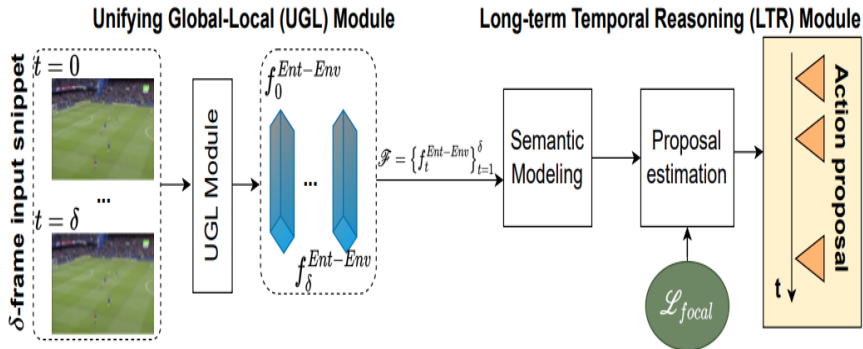
- ▶ Self-attention layer to model interactions
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3. Output: Unified entities and environment representation per frame capturing both background context and key object semantics



2. UGL Architecture

Long-Term Temporal Reasoning (LTR)

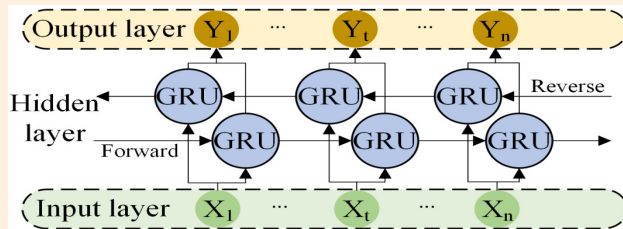


2. UGL Architecture

Long-Term Temporal Reasoning (LTR)

Temporal Modeling

- Input: Sequence of fused frame features for T -frame snippet
- Model: 1-layer **bidirectional GRU** capturing semantic relationships
- Prediction: Per-frame class scores via fully connected + softmax



Liu JH, Yang WH, He J, Wang ZM, Jia L, Zhang CF, Yang WW. Intelligent prediction of rail corrugation evolution trend based on self-attention bidirectional TCN and GRU. *Intell. Robot.* 2024;4(4):318–338.

3. Benchmark Datasets

Three Sports Domains

SoccerNet-v2

- ~550 matches
- ~764 hours of video
- >300k labeled spots
- Strong clutter, camera changes

FineDiving

- 3,000 diving clips
- Phase transitions (entry, twist, etc.)

FineGym

- 5,374 gym performances
- 32 spotting classes
- Converted from action segments to timestamps

4. Quantitative Results

SoccerNet-v2; FineDiving & FineGym

Tabelle: SoccerNet-v2 challenge quantitative comparison (T-mAP).

Method	T-mAP (%)	Gap to UGL (%)	Notes
UGL (ours)	69.38	–	current SOTA
E2E-Spot	66.73	-2.65	previous SOTA
Soares (two-phase)	67.81	-1.57	1st 2022

Tabelle: FineDiving and FineGym improvements of UGL over E2E-Spot (no optical flow).

Dataset	Δ mAP (Gap to E2E-Spot) (%)
FineDiving	87.7 (+2.4)
FineGym	67.8 (+1.3)

- UGL unifies global environment and local entity features to produce precise, interpretable per-frame representations
- Top result on SoccerNet-v2: **69.38% T-mAP** (Top-1); +2.65% vs E2E-Spot
- Consistent gains on FineDiving (+2.4% mAP) and FineGym (+1.3% mAP)
- Adaptive Attention selects relevant entities; GRU + Focal Loss handles temporal reasoning and class imbalance effectively

Thank you for listening!

Questions are welcome